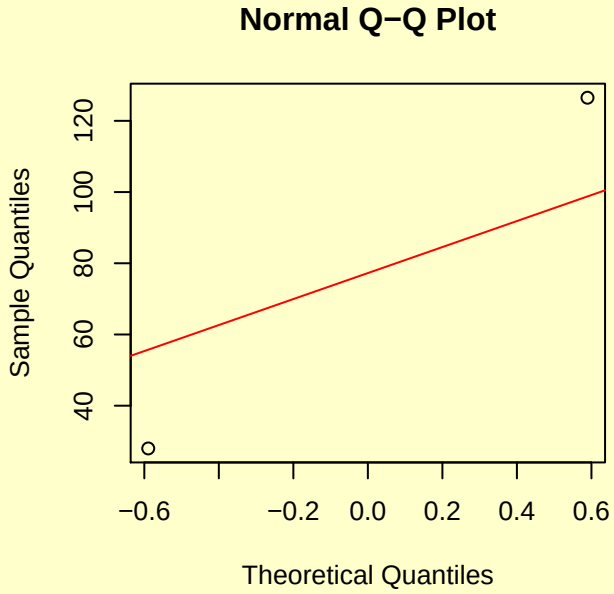
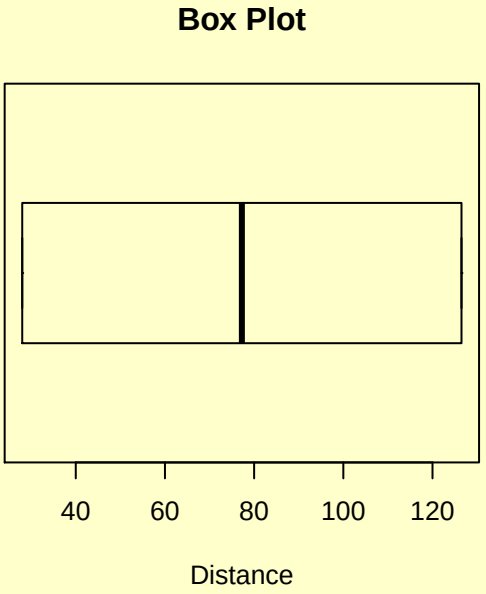
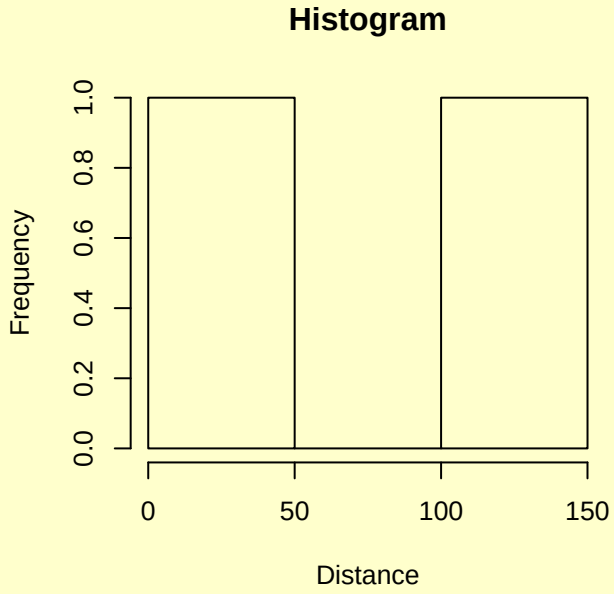
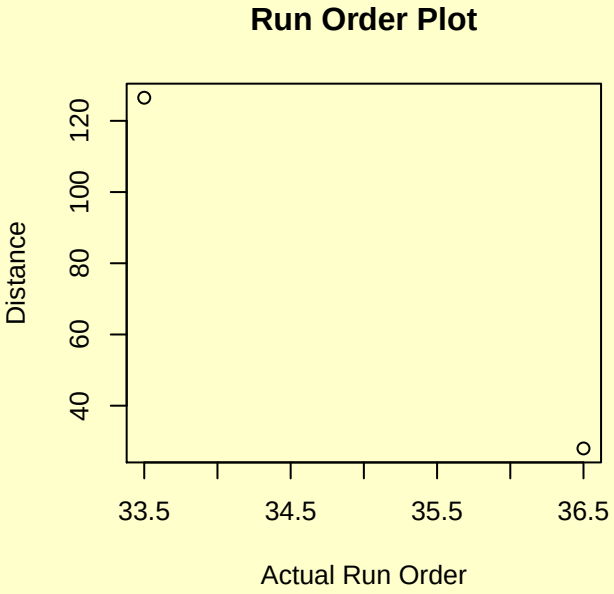
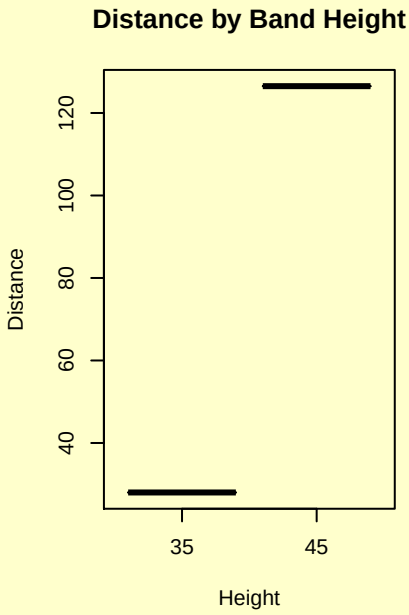
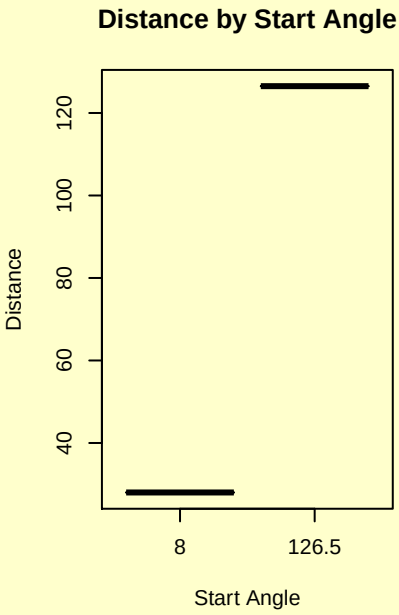

Normal Q−Q Plot
Sample Quantiles
40
60
80
100
120
−0.6
−0.2
0.0
0.2
0.4
0.6
Theoretical Quantiles

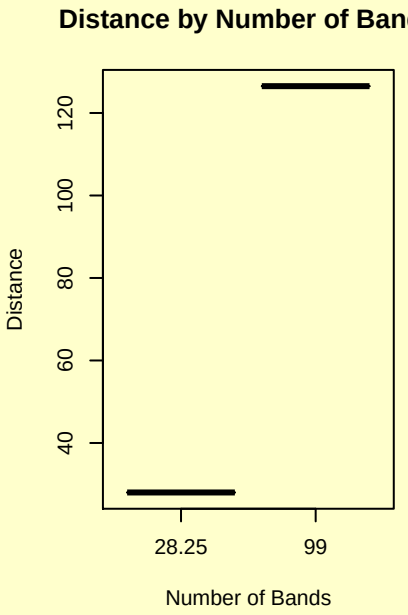

Box Plot
40
60
80
100
120
Distance

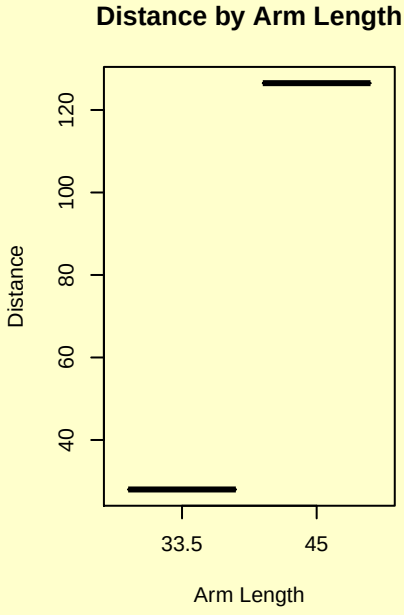

Histogram
Frequency
0.0
0.2
0.4
0.6
0.8
1.0
0
50
100
150
Distance

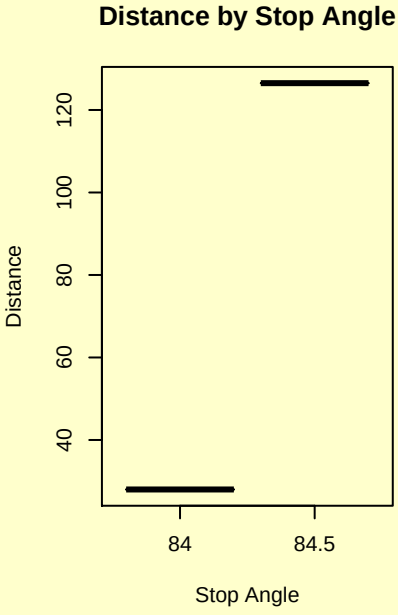

Run Order Plot
Distance
40
60
80
100
120
33.5
34.5
35.5
36.5
Actual Run Order

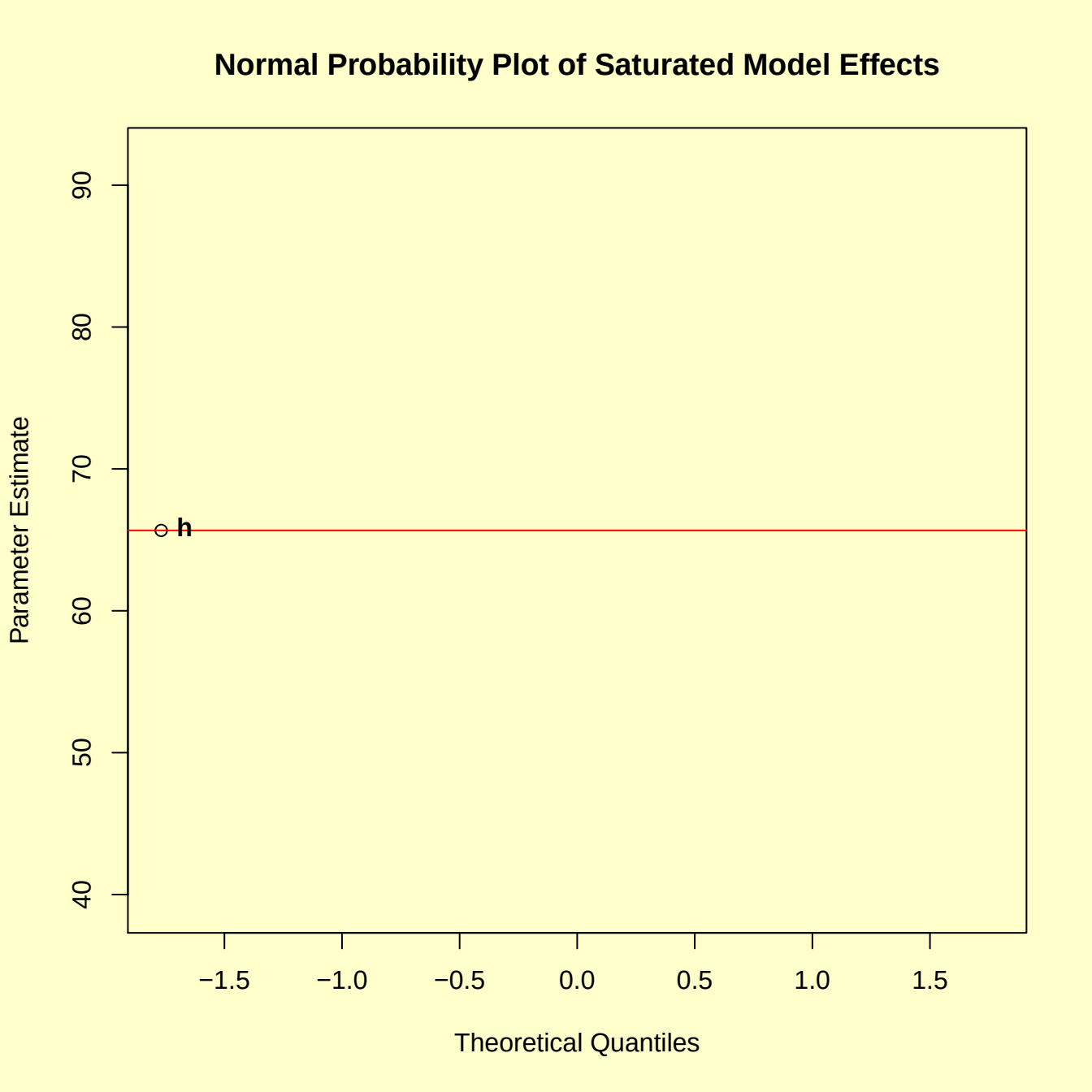

Distance by Band Height
Distance
40
60
80
100
120
35
45
Height

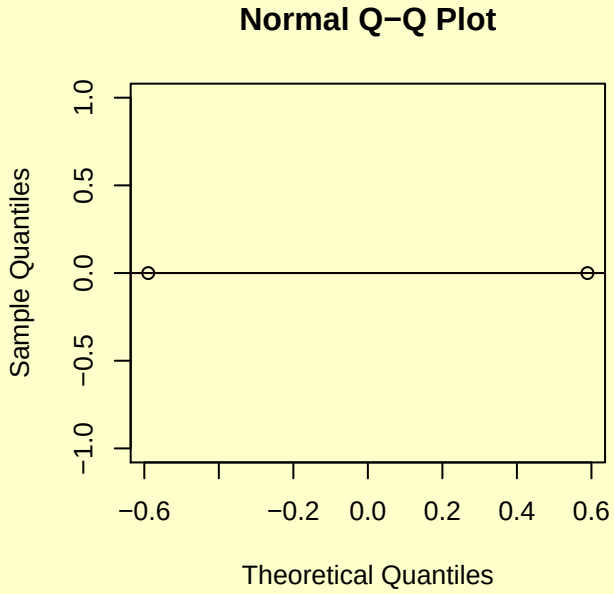

Distance by Start Angle
Distance
40
60
80
100
120
8
126.5
Start Angle

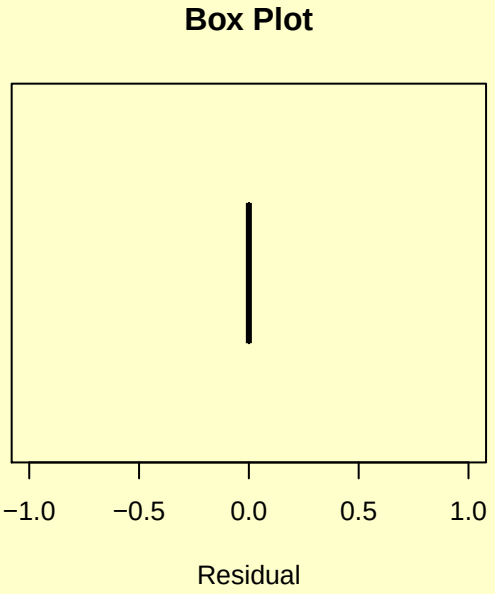

Distance by Number of Bands
Distance
40
60
80
100
120
28.25
99
Number of Bands

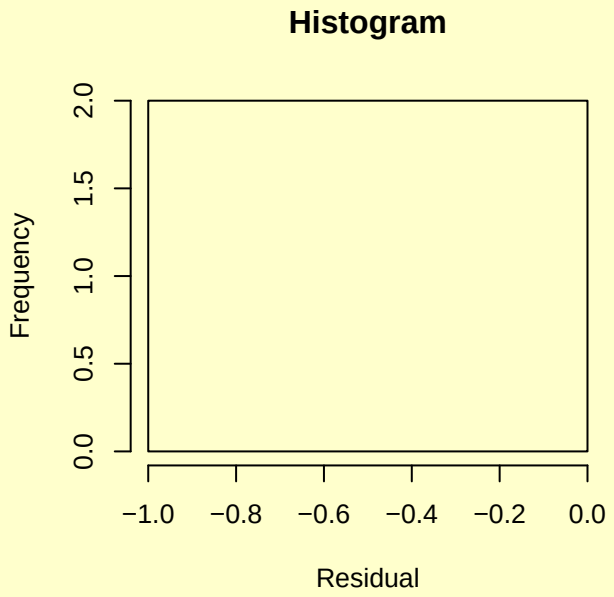

Distance by Arm Length
Distance
40
60
80
100
120
33.5
45
Arm Length

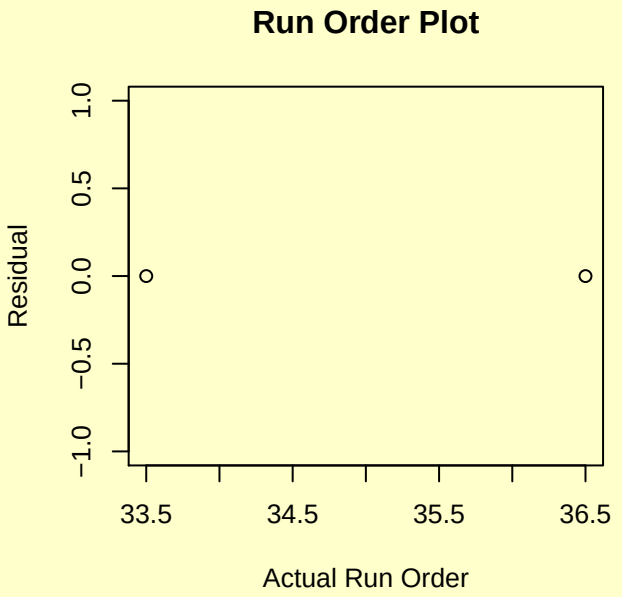

Distance by Stop Angle
Distance
40
60
80
100
120
84
84.5
Stop Angle

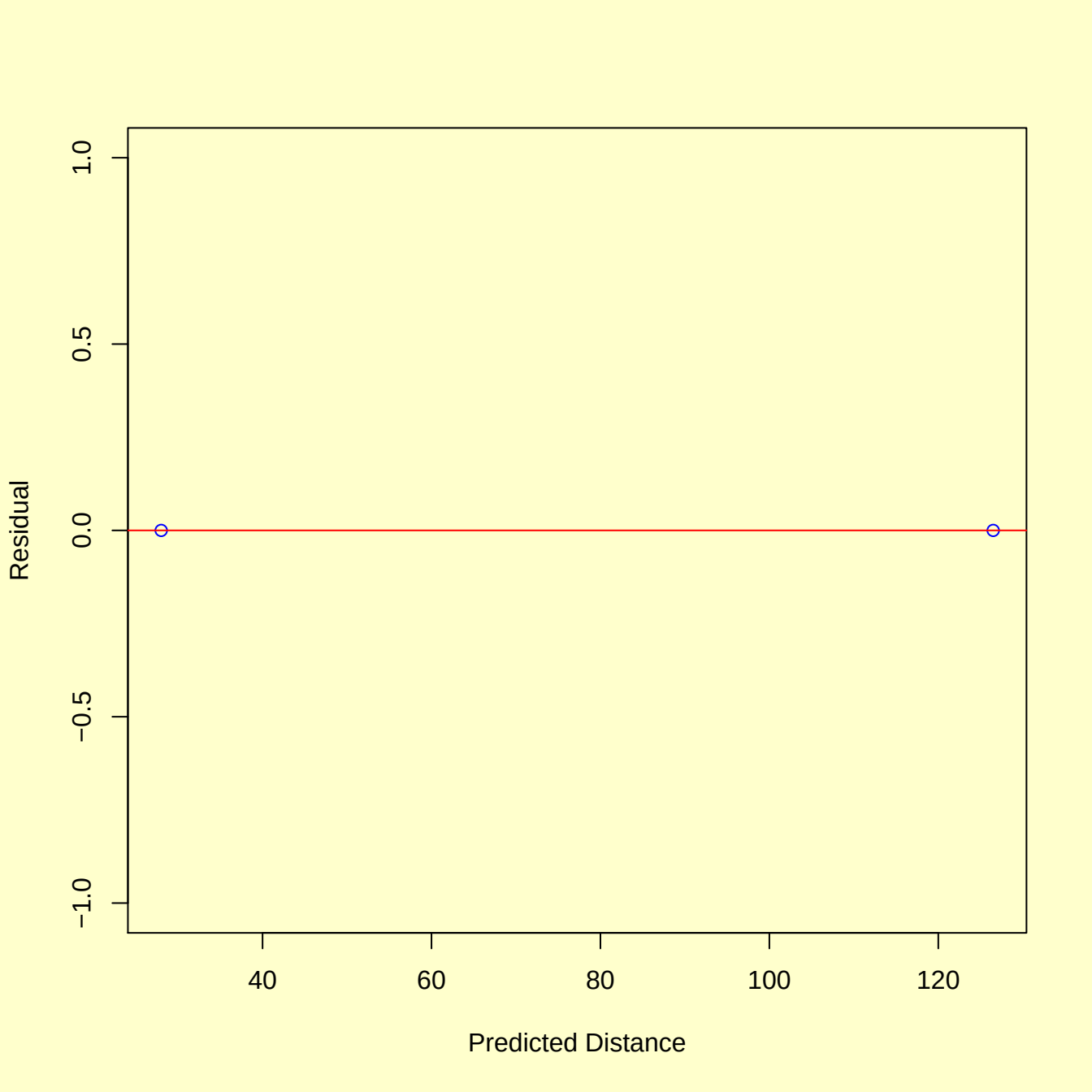

Normal Probability Plot of Saturated Model Effects
Parameter Estimate
90
80
70
60
50
40
h
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
Theoretical Quantiles

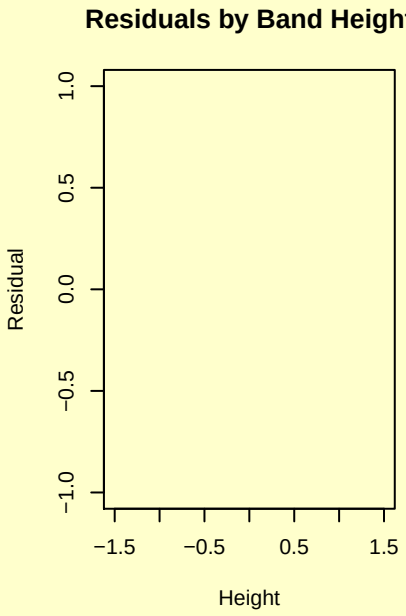

Normal Q−Q Plot
Sample Quantiles
1.0
0.5
0.0
−0.5
−1.0
−0.6
−0.2
0.0
0.2
0.4
0.6
Theoretical Quantiles

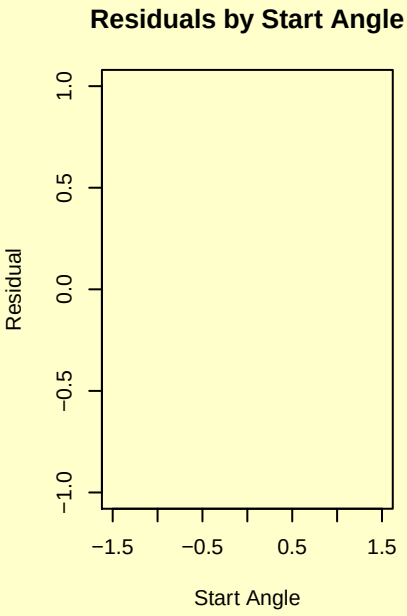

Box Plot
−1.0
−0.5
0.0
0.5
1.0
Residual

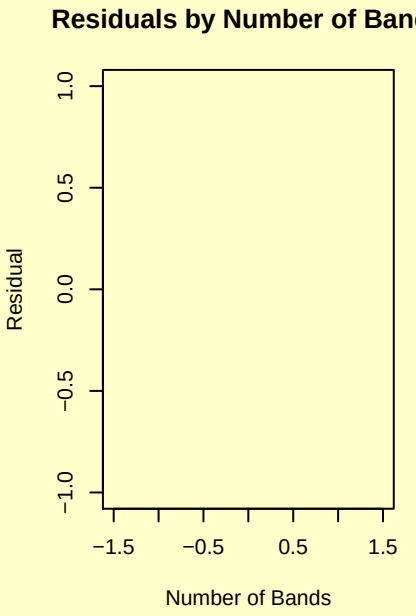

Histogram
Frequency
2.0
1.5
1.0
0.5
0.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
Residual

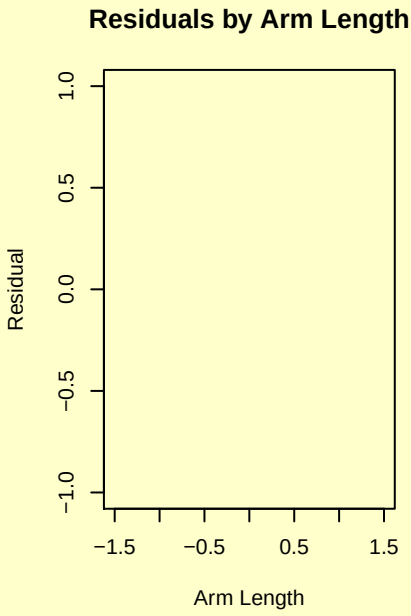

Run Order Plot
Residual
1.0
0.5
0.0
−0.5
−1.0
33.5
34.5
35.5
36.5
Actual Run Order

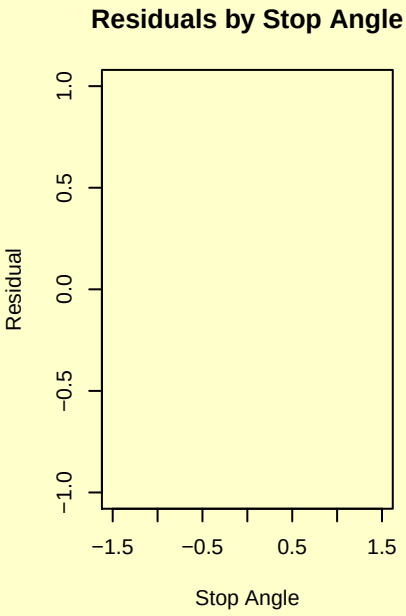

Residual
1.0
0.5
0.0
−0.5
−1.0
40
60
80
100
120
Predicted Distance

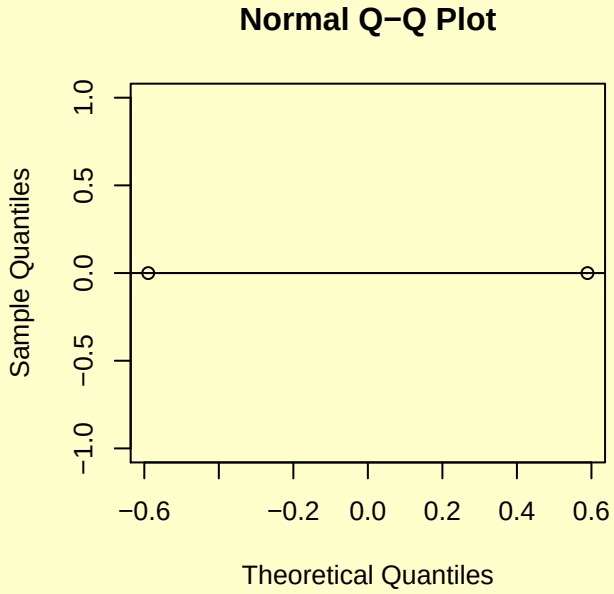

Residuals by Band Height
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Height

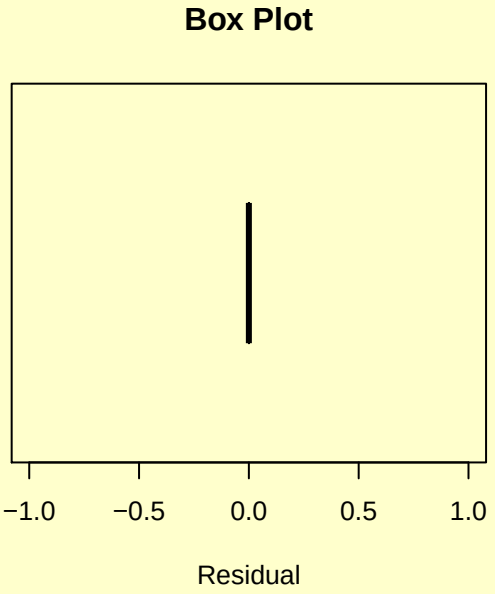

Residuals by Start Angle
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Start Angle

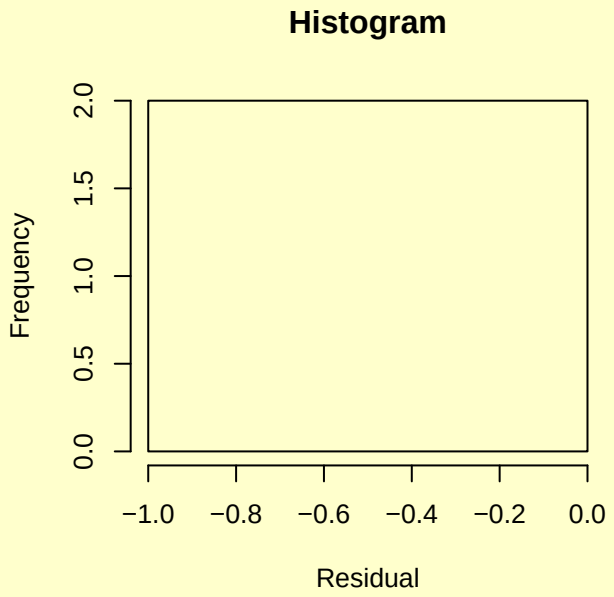

Residuals by Number of Bands
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Number of Bands

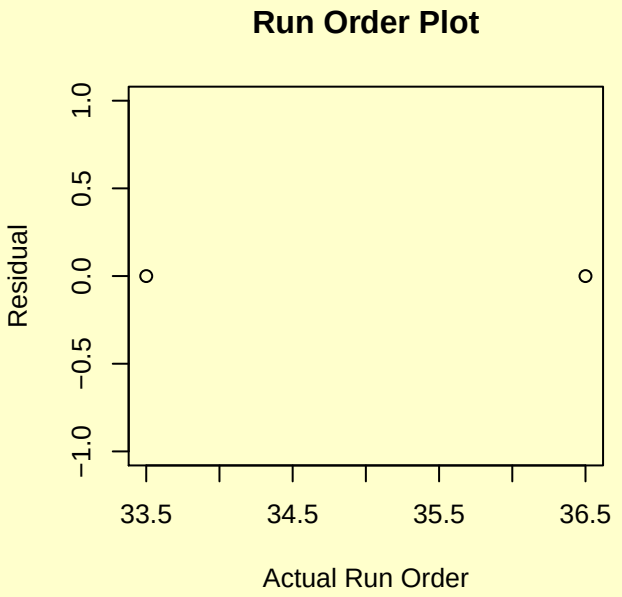

Residuals by Arm Length
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Arm Length

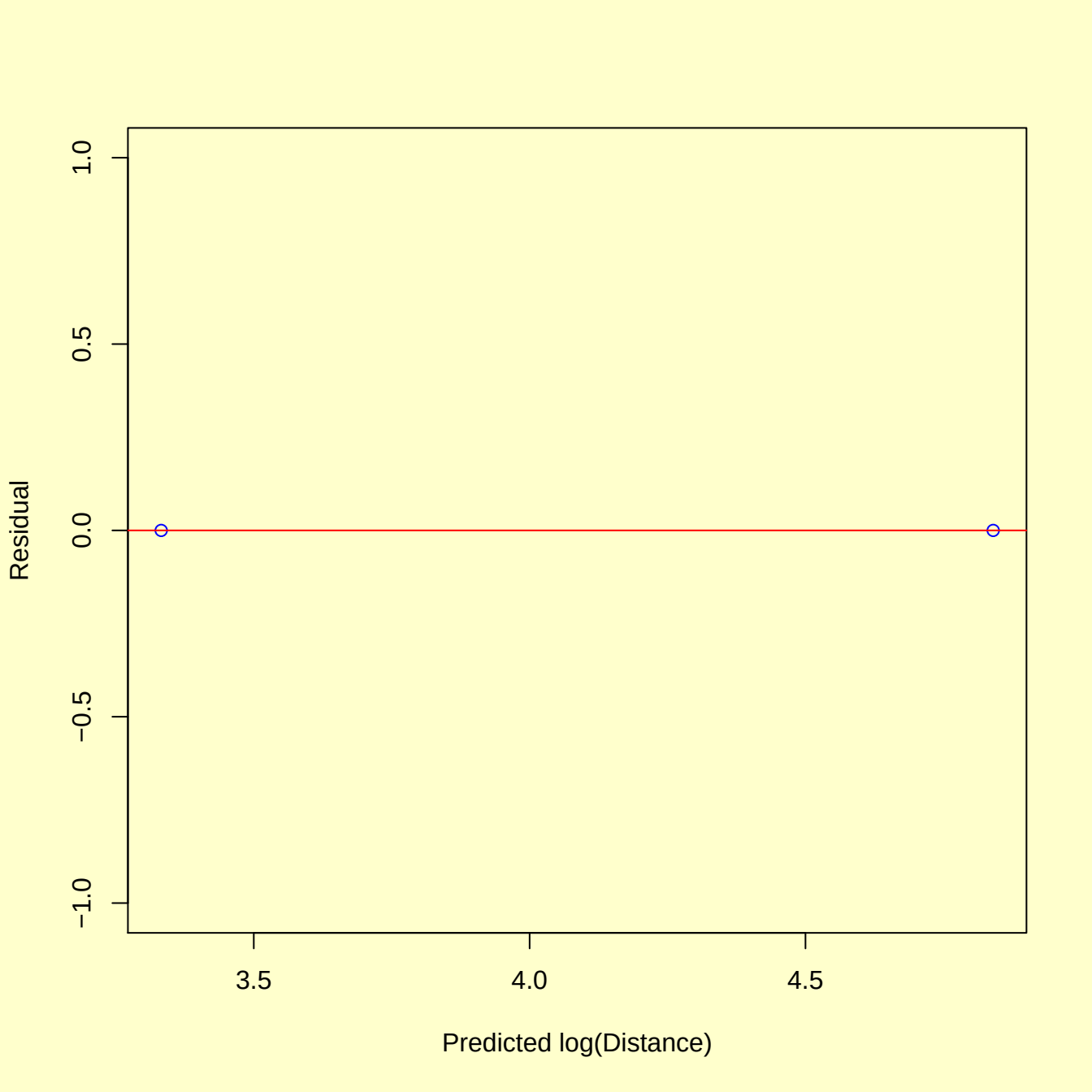

Residuals by Stop Angle
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Stop Angle

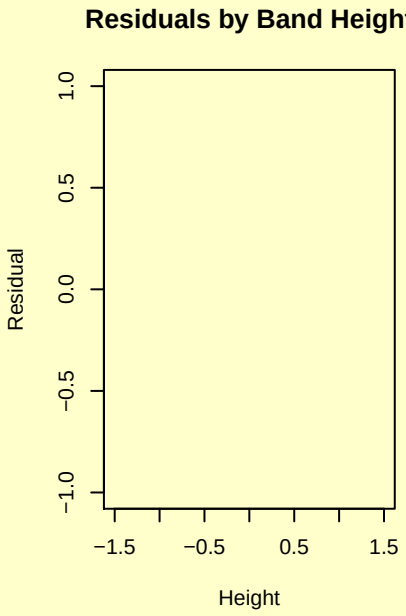

Normal Q−Q Plot
Sample Quantiles
1.0
0.5
0.0
−0.5
−1.0
−0.6
−0.2
0.0
0.2
0.4
0.6
Theoretical Quantiles

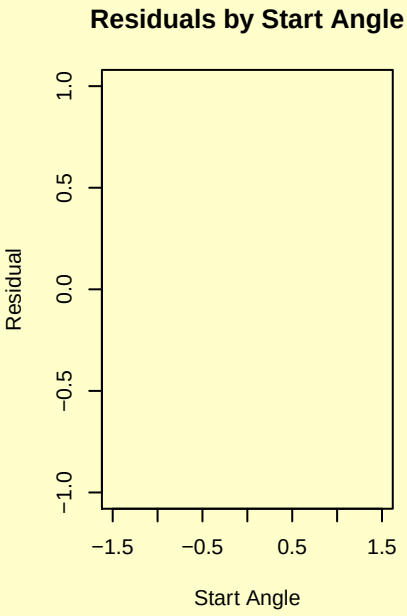

Box Plot
−1.0
−0.5
0.0
0.5
1.0
Residual

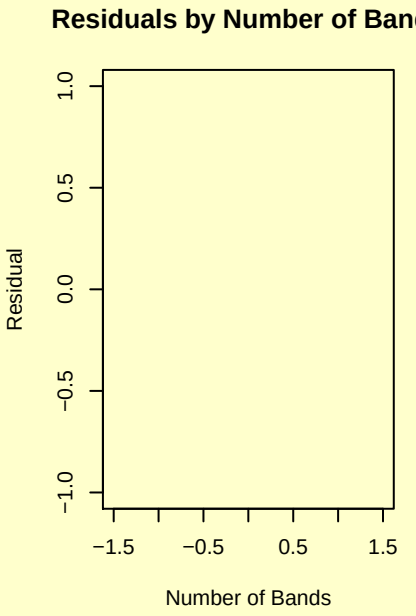

Histogram
Frequency
2.0
1.5
1.0
0.5
0.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
Residual

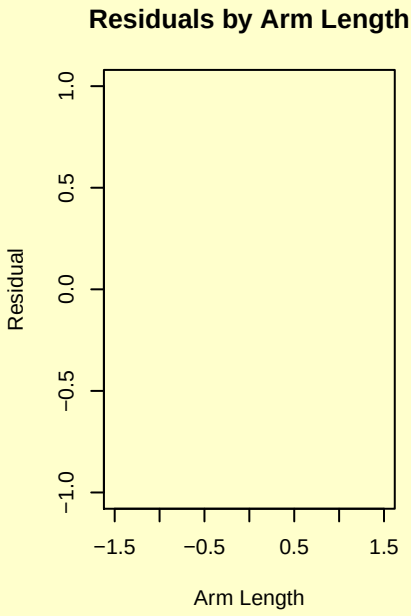

Run Order Plot
Residual
1.0
0.5
0.0
−0.5
−1.0
33.5
34.5
35.5
36.5
Actual Run Order

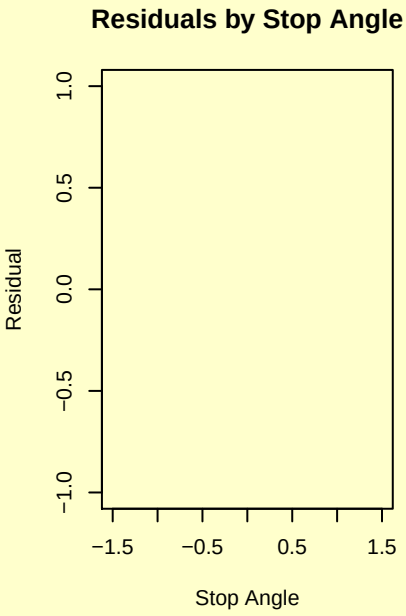

Residual
1.0
0.5
0.0
−0.5
−1.0
3.5
4.0
4.5
Predicted log(Distance)


Residuals by Band Height
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Height


Residuals by Start Angle
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Start Angle


Residuals by Number of Bands
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Number of Bands


Residuals by Arm Length
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Arm Length


Residuals by Stop Angle
Residual
1.0
0.5
0.0
−0.5
−1.0
−1.5
−0.5
0.5
1.5
Stop Angle